

Advanced Linear Algebra: Foundations to Frontiers

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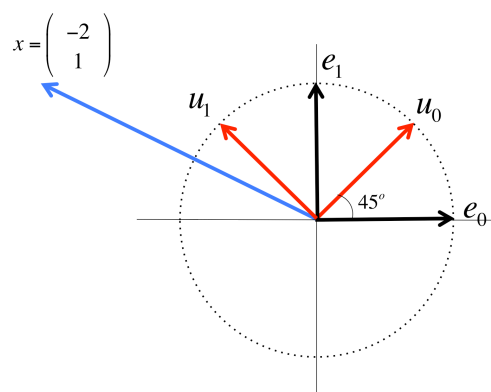
2.2.6 Change of orthonormal basis ¶

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Homework 2.2.6.1. Consider the vector $x = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$ and the following picture that depicts a rotated basis with basis vectors u_0 and u_1 .



What are the coordinates of the vector x in this rotated system? In other words, find $\hat{x} = \begin{pmatrix} \hat{x}_0 \\ \hat{x}_1 \end{pmatrix}$ such that $\hat{x}_0 u_0 + \hat{x}_1 u_1 = x$.

▼ **Solution**

There are a number of approaches to this. One way is to try to remember the formula you may have learned in a pre-calculus course about change of coordinates. Let's instead start by recognizing (from geometry or by applying the Pythagorean Theorem) that

$$u_0 = \begin{pmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{pmatrix} = \frac{\sqrt{2}}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad u_1 = \begin{pmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{pmatrix} = \frac{\sqrt{2}}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Here are two ways in which you can employ what you have discovered in this course:

- Since u_0 and u_1 are orthonormal vectors, you know that

$$\begin{aligned} x &= \begin{matrix} < u_0 \text{ and } u_1 \text{ are orthonormal} > \\ \underbrace{(u_0^T x) u_0} & + & \underbrace{(u_1^T x) u_1} \\ \text{component in the} & & \text{component in the} \\ \text{direction of } u_0 & & \text{direction of } u_1 \end{matrix} \\ &= \begin{matrix} < \text{instantiate } u_0 \text{ and } u_1 > \\ \left(\frac{\sqrt{2}}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}^T \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right) u_0 + \left(\frac{\sqrt{2}}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}^T \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right) u_1 \end{matrix} \\ &= \begin{matrix} < \text{evaluate} > \\ -\frac{\sqrt{2}}{2} u_0 + \frac{3\sqrt{2}}{2} u_1. \end{matrix} \end{aligned}$$

- An alternative way to arrive at the same answer that provides more insight. Let $U = \begin{pmatrix} u_0 & | & u_1 \end{pmatrix}$. Then

$$\begin{aligned}
& x \\
&= \text{< } U \text{ is unitary (or orthogonal since it is real valued) >} \\
& UU^T x \\
&= \text{< instantiate } U \text{ >} \\
&\left(\begin{array}{c|c} u_0 & u_1 \end{array} \right) \begin{pmatrix} \frac{u_0^T}{u_1^T} \end{pmatrix} x \\
&= \text{< matrix-vector multiplication >} \\
&\left(\begin{array}{c|c} u_0 & u_1 \end{array} \right) \begin{pmatrix} \frac{u_0^T x}{u_1^T x} \end{pmatrix} \\
&= \text{< instantiate >} \\
&\left(\begin{array}{c|c} u_0 & u_1 \end{array} \right) \begin{pmatrix} \frac{\sqrt{2}}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}^T \begin{pmatrix} -2 \\ 1 \end{pmatrix} \\ \frac{\sqrt{2}}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}^T \begin{pmatrix} -2 \\ 1 \end{pmatrix} \end{pmatrix} \\
&= \text{< evaluate >} \\
&\left(\begin{array}{c|c} u_0 & u_1 \end{array} \right) \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{3\sqrt{2}}{2} \end{pmatrix} \\
&= \text{< simplify >} \\
&\left(\begin{array}{c|c} u_0 & u_1 \end{array} \right) \begin{pmatrix} \frac{\sqrt{2}}{2} \begin{pmatrix} -1 \\ 3 \end{pmatrix} \end{pmatrix}
\end{aligned}$$

Below we compare side-by-side how to describe a vector x using the standard basis vectors $e_0, \dots, e_{m-1}$ (on the left) and vectors $u_0, \dots, u_{m-1}$ (on the right):

The vector $x = \begin{pmatrix} x_0 \\ \vdots \\ x_{m-1} \end{pmatrix}$ describes

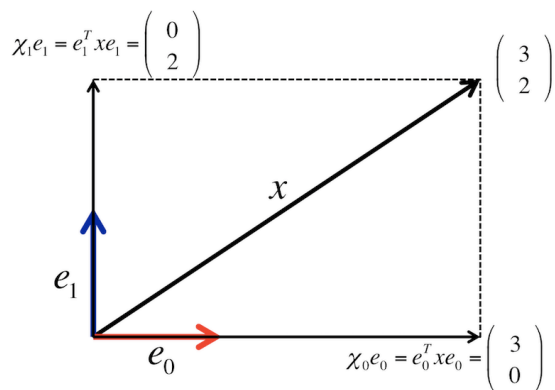
the vector x in terms of the standard basis vectors $e_0, \dots, e_{m-1}$:

The vector $\hat{x} = \begin{pmatrix} u_0^T x \\ \vdots \\ u_{m-1}^T x \end{pmatrix}$ describes

the vector x in terms of the orthonormal basis $u_0, \dots, u_{m-1}$:

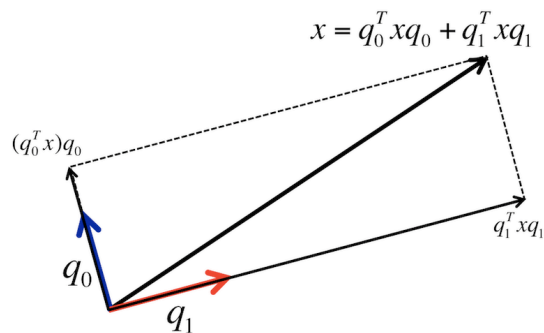
$$\begin{aligned}
x &= \langle x = Ix = IIx = II^T x \rangle \\
II^T x &= \langle \text{expose columns of } I \rangle \\
&\left(\begin{array}{c|c|c} e_0 & \cdots & e_{m-1} \end{array} \right) \begin{pmatrix} \frac{e_0^T}{e_{m-1}^T} \\ \vdots \\ x \end{pmatrix} \\
&= \langle \text{evaluate} \rangle \\
&\left(\begin{array}{c|c|c} e_0 & \cdots & e_{m-1} \end{array} \right) \begin{pmatrix} \frac{e_0^T x}{e_{m-1}^T x} \\ \vdots \\ x \end{pmatrix} \\
&= \langle e_j^T x = \chi_j \rangle \\
&\left(\begin{array}{c|c|c} e_0 & \cdots & e_{m-1} \end{array} \right) \begin{pmatrix} \frac{\chi_0}{\chi_{m-1}} \\ \vdots \\ x \end{pmatrix} \\
&= \langle \text{evaluate} \rangle \\
&\chi_0 e_0 + \chi_1 e_1 + \cdots + \chi_{m-1} e_{m-1}.
\end{aligned}$$

Illustration:



$$\begin{aligned}
x &= \langle x = Ix = UU^H x \rangle \\
UU^H x &= \langle \text{expose columns of } U \rangle \\
&\left(\begin{array}{c|c|c} u_0 & \cdots & u_{m-1} \end{array} \right) \begin{pmatrix} \frac{u_0^H}{u_{m-1}^H} \\ \vdots \\ x \end{pmatrix} \\
&= \langle \text{evaluate} \rangle \\
&\left(\begin{array}{c|c|c} u_0 & \cdots & u_{m-1} \end{array} \right) \begin{pmatrix} \frac{u_0^H x}{u_{m-1}^H x} \\ \vdots \\ x \end{pmatrix} \\
&= \langle \text{evaluate} \rangle \\
&u_0^H x u_0 + u_1^H x u_1 + \cdots + u_{m-1}^H x u_{m-1}.
\end{aligned}$$

Illustration (q should be u here):



Another way of looking at this is that if $u_0, u_1, \dots, u_{m-1}$ is an orthonormal basis for $\mathbb{C}^m$, then any $x \in \mathbb{C}^m$ can be written as a linear combination of these vectors:

$$x = \alpha_0 u_0 + \alpha_1 u_1 + \cdots + \alpha_{m-1} u_{m-1}.$$

Now,

$$\begin{aligned}
u_i^H x &= u_i^H (\alpha_0 u_0 + \alpha_1 u_1 + \cdots + \alpha_{i-1} u_{i-1} + \alpha_i u_i + \alpha_{i+1} u_{i+1} + \cdots + \alpha_{m-1} u_{m-1}) \\
&= \alpha_0 \underbrace{u_i^H u_0}_0 + \alpha_1 \underbrace{u_i^H u_1}_0 + \cdots + \alpha_{i-1} \underbrace{u_i^H u_{i-1}}_0 \\
&\quad + \alpha_i \underbrace{u_i^H u_i}_1 + \alpha_{i+1} \underbrace{u_i^H u_{i+1}}_0 + \cdots + \alpha_{m-1} \underbrace{u_i^H u_{m-1}}_0 \\
&= \alpha_i.
\end{aligned}$$

Thus $u_i^H x = \alpha_i$, the coefficient that multiplies u_i .

Remark 2.2.6.1. The point is that given vector x and unitary matrix U , $U^H x$ computes the coefficients for the orthonormal basis consisting of the columns of matrix U . Unitary matrices allow one to elegantly change between orthonormal bases.